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MINI REVIEW

Biomimetically precipitated nanocrystalline hydroxyapatite

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ABSTRACT

Nanocrystalline hydroxyapatite has been extensively used for biomedical field, such as drug carrier, antitumor agent, surface coating, and bone-regenerating material. The wet chemical method is one of the most commonly used methods in the synthesis of nanosized hydroxyapatite due to its relatively low costs and more customizable control of the final product. Herein, we updated the most recent advances in artificial nano-hydroxyapatite prepared from biomimetic precipitation throughout context. Furthermore, micron-sized particles and macro-sized scaffolds made of nano-hydroxyapatite deposition are introduced. Their outstanding physicochemical properties and potential clinical application are highlighted. This article reviews the latest progress on preparing biomimetic nano-hydroxyapatite and provides inspiration to promote new breakthrough in regenerative therapy and clinical translation.

KEYWORDS

nano-hydroxyapatite, biomimetic, biomedical application, biofabrication

1 Introduction

Regeneration of bone defects caused by injury, infection, tumor excision, and age-related diseases is a clinical challenge. In order to reconstruct the deficient bone for improving the life quality of patients, many bone augmentation techniques using different graft materials have been proposed. Allografts and xenografts from bone banks or other animal species remain the most common strategy to support and stimulate bone growth [1]. Nevertheless, they are still subject to shortage of supply, attenuated osteointegration and disease transmission [1]. Synthetic hydroxyapatite (HAP) can overcome these limitations due to its good biocompatibility, bioactivity, adequate mechanical properties, and a composition identical to bone mineral. Especially, nano-sized HAP (nano-HAP) has been widely used in clinical applications in the form of implant surface coatings, the composition of cements, and scaffolds [2, 3].

HAP at nanoscales present peculiar properties, such as reactivate surface, mechanical properties, electrochemical properties [4]. It is strong surface energy and positive charge of nano-HAP enable them attach to the negative charge surface of tumor cells, preventing tumor cells adhesion and proliferation [5]. Nano-HAP has been used as an adsorbent due to the high surface area and surface functional groups (both positively charged Ca^{2+} and negatively charged PO_4^{3-} ions) [6]. Serum proteins would be efficiently adsorbed on the surface of nano-HAP, which are crucial for mediating cell attachment. Moreover, nano-HAP can influence the tensile strength of gel composites by mixing different nano-HAP content [7]. In particular, in the review of He et al., association between physical properties of nano-HAP and tumor cell response are accurately described [5]. Thus, different from conventional micro-sized HAP, nano-HAP can be served as regenerative medicine, anti-tumor agents [8], efficient drug delivery cargos [2], few environmental applications [6].

Although, a great number of literatures have been reported and discussed various synthesis routes to obtain nano-HAPs with controlled shape and size, but solid-state methods and wet chemical methods have been the most common used for nano-HAP produced. The first method involves milling and calcination to obtain nanopowder from HAP precursors (biochemical reagents or biowastes) [9–11]. Thermal process processes and corrosive reagents were inevitably used in processing bio-samples for removing the impurities. This mechanochemical process is time-consuming, highly energy

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consumption and the products are irregular shapes with heterogeneous phase composition [9, 12]. Different from the natural nano-HAP synthesis routes, chemical methods are more tunable in size and morphology of nano-HAP. In recent years, several chemical synthesis methods have been identified: hydrothermal, sol-gel, co-precipitation, and biomimetic deposition methods [13]. However, these methods except for biomimetic methods require harsh reaction equipment and conditions, such as alkaline medium, high temperature, unsafety surfactants, and microwave irradiation, in order to produce nano-HAP with properties like biological apatite [13].

In the physiological process, nanosized plates-like HAP crystals grow in pre-organized type I collagen which controls the nucleation and orientation of HAP nanoparticles to the long axis of the collagen fibrils [14]. Bone mineralization undergoes self-assembly and mineral maturation storing ions as a reservoir and constructing a hierarchical structure for bone and teeth [14]. Inspired by the concept of the formation of natural nano-HAP in hard tissue, researchers have been made great advances in the preparation process of biomimetic nano-HAP. However, few papers emphasized the biomimetic deposition technique for preparing nano-HAP.

Here, we aim to review the biomimetic synthesis of nano-HAP and provide practical routines to mimic the process of biomineralization and remineralization of nano-HAP. The clinical application and translation concerns are also covered in the present review.

2 Preparation

Biomimetic preparation of nano-HAP differed from conventional wet chemical methods in two ways: mineralization deposition of saturated solutions and remineralization of bio-macromolecule. Stimulated body fluid (SBF) was initially introduced by Kokubo et al. to produce bio-glass with a biomimetic apatite coating [15]. The concentrations and composition of ions in SBF nearly were equal to those of human blood plasma [16]. So it is a common method to predict *in vivo* bone bioactivity by immersing biomaterials in SBF. On the other hand, the biomimetic calcium phosphate coating technique was also developed by sequential immersion in saturated solution (supersaturated SBF and Ca-P solution) [17]. This coating could be potentially served as a drug carrier deposited with the layer of amorphous calcium phosphate and the octacalcium phosphate under a physiological condition. Several research groups were inspired by this tech for producing nano-HAP [18, 19]. Macromolecules like BSA, collagen fiber, and other biopolymers could provide pre-organized matrices for modulating nucleation and oriented growth of nano-HAP [20–23]. It would raise the concern about how to separate these organics to get pure nano-HAP. Calcination is a direct method to remove unwanted impurities but at high costs [20]. In other cases, separating nano-HAP from its collagen composites is unnecessary because of its biomimetic constructs like native bone. Cui and his team developed mineralized collagen composites for clinical applications [22]. It is a remineralization process with biomacromolecule that is prepared from saturated Ca-P solution mixed with redissolved collagen by adjusting pH to 7. The main component was not nano-HAP but collagen (50–55 wt%) which result in less mechanical stability and high degradation rate [24]. Here, we discussed the main points in preparing biomimetic nano-HAP as stated in Fig. 1.

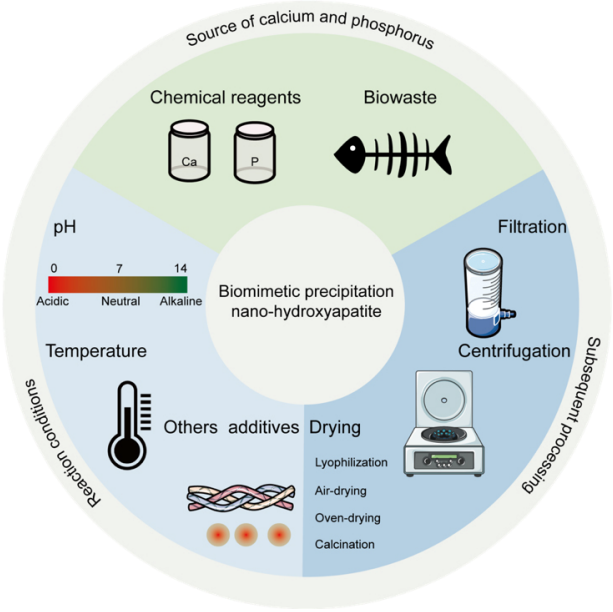

Figure 1 Key factors for biomimetic preparation of nano-HAP.

2.1 Source of calcium and phosphorus

Nano-HAp can be directly extracted from natural resources (bovine bone, fish bones, eggshells, etc.) or prepared calcium precursor (such as CaO) from biowastes [25–27]. The utilization of this waste in producing HAP preserved trace elements within bone minerals which are necessary ions (such as Mg^{2+}, Sr^{2+}, Zn^{2+}, K^{+}, and Na^{+}) that could accelerate the bone regeneration process, osseointegration [27]. However, the calcium source extracted from mammalian sources is HAP but shell sources, such as eggshells, snail shells, and mussel shells, are almost composed of calcium carbonate [27]. Further processing is required to remove protein or transform mineral phase which means cannot be used directly for HAP production. Moreover, whether the synthesis of nano-HAP from natural raw materials through mechanochemical method or chemical precipitation method, it would raise the concern of potential infection, the transmission of disease, and heavy metal pollution [27]. Various phosphate precursors, such as $CaHPO_4$, $Ca_3(PO_4)_2$, Na_3PO_4, $(NH_4)_2HPO_4$, $NH_4H_2PO_4$, and H_3PO_4, are used in the production of HAP. Boonyang et al. compared three phosphate precursors (H_3PO_4, $Ca_3(PO_4)_2$, and $(NH_4)_2HPO_4$) reacted with calcium carbonate from crocodile eggshells for hydrothermal reaction [28]. The results demonstrated that the phosphate compound that gave more amount of phosphate ions would prepare HAP more efficiently [28]. Ishtiaque et al. prepared nano-HAP by wet chemical precipitation and concluded that starting materials would influence the thermal stability and hardness of HAP [29]. To this end, commercially available chemical reagents would be used as an alternative by dissolving in a water solution to generate a uniform precipitate and a high-purity nano-powder material would be obtained after post-reaction processing.

2.2 Reaction conditions

The synthesis conditions were varied in the process of wet chemical precipitation, especially in biomimetic preparation (Table 1). A series of synthesis conditions were investigated by Wang et al. and provided evidence that both pH value and reaction temperature played important roles in the morphology

Table 1 Biomimetic precipitation method for the synthesis of Nano-HAP

NO.	Mineral source	Macromolecular Additives	Reaction conditions	Subsequent processing	Finding	Ref.
1	$Ca(NO_3)_2$ $(NH_3)_2HPO_4$	BSA	pH: 9–11 30 ± 2 °C	Spray dried Sintered at 1,000, 1,200, 1,300 °C	HAP powder of 30–40 nm with acicular morphology	[20]
2	$CaCl_2$ NaH_2PO_4	Recombinant human procollagen	pH: 7 25 °C	centrifugation freeze-dried	Nano-rods with their c-axis parallel to the Collagen fibrils	[23]
3	$Ca(NO_3)_2$ $(NH_3)_2HPO_4$	PVA and BSA	pH: alkaline 75 °C	Washing	Nanosized HAP grew on mineralized gels	[21]
4	$Ca(CH_3COO)_2$ $NH_4H_2PO_4$ $(NH_4)_2HPO_4$	Agar and Collagen I	pH: 6.5, 7.0, and 7.5 37 °C	Centrifugation	Collagen and pH modulated biomineralization	[34]
5	$CaCl_2$ NaH_2PO_4 SBF	—	pH: 7.25 37 °C	Drying at 80 °C for 24 h Calcination at 900 °C for 2 h	Shorter precipitation times	[18]
6	$CaCl_2$ K_2HPO_4 SBF CaP-Tris solution	—	pH: neutral 37 °C	Centrifugation sintering at 700 °C	HA nanoparticles were synthesized using CaP-Tris solution	[19]
7	CaP-Tris solution	—	pH: 7.4 37 °C	Air-drying overnight at 25 °C	HAP granules with internally incorporated BMP2	[45]
8	$Ca(CH_3COO)_2$ H_3PO_4	BSA	pH: 7.5–9.5	Filtered and washed	The BSA release characteristics of BSA-loaded CDHAP	[36]
9	$Ca(NO_3)_2$ Na_3PO_4	Gelatin	pH: 10 25 °C	Filtered and washed; dried overnight at 25 °C	Gelatin as a template agent for nano-HAP	[68]

Abbreviation: SBF, simulated body fluid; CaP-Tris, calcium phosphor Tris solution; PVA, poly(vinyl alcohol); BSA, bovine serum albumin; CDHAP, calcium-deficient hydroxyapatite; BMP2, bone morphogenetic protein-2; 25 °C: room temperature.

of nano-HAP [30]. It suggested that pH value and temperature were predominant conditions in controlling crystal morphology and particle size of nano-HAP. Vilela conducted an experiment to verify the effect of temperature and pH on calcium phosphate precipitation under highly supersaturated conditions [31]. It proved that HAP is formed at pH 6.5 and 45 °C while dicalcium phosphate dihydrate under more acidic or drifting pH [31]. Most nano-HAPs or ion substituted nano-HAP from chemical precipitation react in an alkaline media (usually adjusting to pH 10 by ammonia or sodium hydroxide solution) and higher reaction temperature to ensure the phase purity [8, 31–33]. Furthermore, it is worth noting that macromolecular additives (such as type I collagen) also played a vital role in controlling the crystallization of HAP during the biomineralization process [34]. On the other hand, the aging time ensured whether the reaction was complete or not. The TEM observations from Feng et al. showed the nano-HAP powers increase with extended aging time [35]. This suggests that aging time is another critical factor contributing to crystal growth and agglomeration.

How can we obtain biomimetic nano-HAP precipitate by regulating chemical precipitation reaction conditions? Firstly, nano-HAP was prepared in supersaturated calcium solutions under biomimetic conditions (37 °C and pH 7.4). SBF is similar to the natural medium that is responsible for biological bone formation and commonly used as a growth medium to facilitate the spontaneous nucleation of nano-HAP. Calcium or phosphorus solution coincide in high concentrations of SBF and form a precipitate at physiological pH and temperature. The reaction time reported in most literature was longer than 24 h but Leena et al. accelerated this processing time to a shorter duration (3 h) [18]. These reaction conditions would present a precursor phase (amorphous calcium phosphate or octacalcium phosphate) in the solution and then hydrolyze for thermally stable HAP [33]. This tech was first used in situ coating of biomaterials and refined in producing nano-HAP powders [17]. The produced nanopowders may be in non-stoichiometric HAP with a Ca to P ratio: less than 1.67 (specifically, in the form of calcium-deficient hydroxyapatite) [19]. This might be attributed to that both the calcium and phosphorus precursors in this reaction are not fully reacted and resorbable phases of calcium phosphates were formed. Moreover, very little amount of carbonate would be incorporated in nano-HAP which might be the result of the absorption of atmospheric CO_2 on the surface of HAP particles. However, Paz et al. carried out a different way in which HAP nanoparticles with high purity were prepared in slightly acidic solutions (a maximum of pH: 6.52) [33]. It is sodium bicarbonate that served as a reaction medium making ion concentration sufficient for mineralization [33].

Secondly, another synthesized route is to mimic the process of biomineralization with the help of biomolecules. Previous studies have shown that type I collagen mineralized by CaP nanocrystals resembled the hierarchical structures of lamellar bone in many mammals, including humans. BSA and other macromolecular polymers were also investigated for biomedical nano-HAP. These results have shown that the nanocrystal growth and shape are susceptible to these biomolecules, in which a Ca-bridge with amino acid residues or other anionic groups was built [36, 37]. Here, we shed light on collagen-assisted nano-HAP deposition. Co-precipitation of collagen fibrils and nano-HAP was spontaneous and self-assembled under mild conditions (Fig. 2(b)). Typically, type I collagen is pre-dissolved in PO_4^{3-} containing solution and then mixed with a Ca^{2+} containing solution to allow mineral deposition. Zhai et al. followed this procedure and mineralized recombinant human-like collagen with the formation of nano-HAP crystals on the fibrils [23]. The nano-HAP crystals embedded in

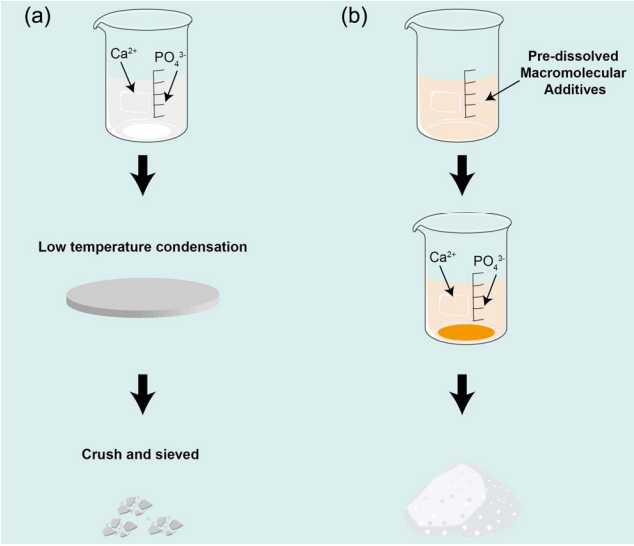

Figure 2 The schematic diagram of the biomimetically precipitated nano-HAP procedure with two routes. (a) Preparation for precipitated nanocrystalline calcium phosphate particles; (b) Macromolecular-aided preparation for biomimetic HAP powders.

collagen matrix with the c-axis of HAP parallel to the long axis direction of collagen fibrils as is shown by TEM morphology and selected area electron diffraction (SAED) [23]. Although the matrix-mediated biomineralization can produce nano-HAP composites that mimic natural bone to some extent, nano-HAP produced in this way is at higher costs and lower productivity when compared with other wet chemical precipitation methods. Furthermore, the nano-HAP/collagen composite is lack mechanical properties (mechanical rigidity, hardness, and load-bearing strength). Typically, it is strengthened by biomedical polymer when confronted with large bone defects [24]. The usage of BSA as a crystal nuclear initiator and inhibitor would be better for manufacturing nano-HAP since BSA is most commonly available and stable up to 78 °C in the presence of a high ionic concentration [20].

To sum up, biomimetic nano-HAP can be prepared from two routes which need to organize CaP substances and reaction conditions. There are two advantages over the wet chemical methods to produce nano-HAP: (1) In the process of biomineralization, the inorganic constituent ions, like Na^+, K^+, Mg^{2+}, and CO_3^{2-}, from saturated ionic solution (such as simulated body fluid solution or calcium phosphor tris solution) could be preserved in the precipitate without other ionic additives; (2) remineralization with biomacromolecule triggers self-assembly and guides crystal growth like nano-HAP/collagen nanocomposites with a hierarchical structure closer to that of bone-liked apatite. The two routes make incorporation of biological compounds and bulk synthesis of nano-HAP possible due to the high safety of reaction conditions (pharmaceutical grade starting materials under the physiological condition without the use of potentially harmful surfactants or templating agents). Thus, by introducing biomimetic preparation, the biological performance and clinical application of the nano-HAP would be further improved.

2.3 Subsequent processing of the final product

To remove excess reactants in the precipitation and obtain purer nano-HAP, the subsequent processing of filtration, centrifugation, dialysis, and washing are used [18, 19, 38]. For example, the unintended and soluble calcium salt, sodium salt, magnesium salt, and organic solvents would retent in sediments which may compromise the biocompatibility of the nanoparticles. After solid-liquid separation, it is urgent to obtain pure and good crystallinity HAP nanoparticles. Drying methods, including but not limited to lyophilization, air-drying, oven-drying, and calcination, are used to remove moisture and improve crystallinity. The drying method of lyophilization freeze nano-HAP at very low temperatures (about –70 °C) and extracts the liquid by sublimation, whereas air-drying (25 °C) or oven-drying (50–110 °C) by air flow and evaporation [18, 33, 39]. Rather broad peaks and H_2O adsorption peaks are observed in the XRD and FTIR, which indicate poor crystallinity and the presence of water [18, 33].

The heat treatment of calcination on nano-HAP precipitates can cause dehydroxylation and possible decomposition, leading to the change in physicochemical properties of the final materials. Leena and her co-worker compared 80 °C dried nano-HAP samples and calcinated samples at 900 °C using the XRD to determine the phase purity and crystallographic analysis [18]. Their results confirmed calcination makes nano-HAP increased crystallinity presenting the XRD patterns with sharper diffraction peaks than 80 °C dried nano-HAP samples. In the FTIR spectrum, a broad peak related to H_2O adsorption was visible around 3,250–3,500 cm^{-1} for 80 °C dried samples while disappeared when the samples were calcinated at 900 °C. The absorption peak of carbonate in the range of 1,420–1,455 cm^{-1} and at 873 cm cm^{-1} is also eliminated in calcinated samples. Thermogravimetric analysis indicated less thermal stability even at high temperatures as well as phase purity when precipitating without calcination. The subsequent heat treatment is an effective procedure to purify the crystal phase, improve thermal stability, remove the adsorbed water, and remove other unwanted additives [18, 20]. However, it will undergo decomposition when nano-HAP calcinated at high temperature (over 900 °C) which would make HAP into tri-calcium phosphate and calcium oxide [18, 20]. And the remineralization of nano-HAP on polymeric substances and nano-HAP internally-incorporated biologicals are not suitable for calcination or heat-treatment over physiological temperature.

3 Characteristics of Nano-HAP

3.1 Physical characteristics

Nanocrystalline HAP in the natural bone is calcium deficient HAP (CDHAP, Ca/P molar ratio is less than 1.67) with different ion substitutions and they are in plate-like or rod-like morphology that is 20–80 nm long and 2–5 nm wide [6, 13]. The Ca/P molar ratio of pure HA is 1.67 with the chemical formula $Ca_{10}(PO_4)_6(OH)_2$, whereas for enamel and dentin, it is 1.62 and 1.64, respectively. The Ca/P ratio for bone mineral of human skeleton varied range from 1.37 to 1.87. This indicates that additional ions, such as sodium, magnesium, zinc, strontium, fluoride, and carbonate, etc. can occur in the unit cell of HAP for the calcium ions, the phosphate groups or the hydroxyl groups. The most abundant is represented by carbonate ion (CO_3^{2-}) in the amount of 3–8 wt%. As a result, the calcium deficient HAP from human bone is structurally incomplete, making it more bioactive and more soluble in body fluids [40, 41].

3.2 Biological response

For safety concerns, Wang et al. co-cultured MC3T3-E1 with different concentrations of nano-HAP (20–30 nm rod-shaped) and investigated the toxicological effect *in vivo* by injecting intra-peritoneally SD rats with nano-HAP particles. It revealed that the nanomaterial of HAP could reduce cell viability and induce apoptosis both *in vitro* and *in vivo*. Zhao et al. discussed different shapes of nano-HAP on cytotoxicity and concluded that needle- or plate-like shape induced more cell death than rod- or sphere-like shape when culture with BEAS-2B cells for 24 h at the concentration of 100 μg·mL^{-1} [42]. As for bioactive efficiency, the size, culture concentration, and morphology of nano-HAP would influence bioactive efficiency. Nano-HAP exerted the inhibition effect on cancer cells while micro-sized HAP did not [8]. Cai et al. found that 20 nm sized HAP nanoparticles were found to proliferate rabbit MSCs [43]. HAP nanorods with aspect ratios showed different osteogenic capacities on rat bone marrow-derived mesenchymal stem cells [44]. 100 μg·mL^{-1} well-dispersed nano-HAP could stimulate osteoblast differentiation [38]. In the end, the toxicity and bioactivity of nano-HAP lie in diverse aspects: dose, aggregation, particle size, morphology, and cell types.

4 Technological evolutions: nano-HAP aggregation-based biobone granule

Our group has achieved nano-HAP deposition under the physiological conditions of temperature (37 °C) and pH (7.4) [45]. The biobone granule was successfully fabricated and intended for clinical translation (Fig. 3). This granule was the aggregation of nanosized HAP particles through apatite-like precipitate drying at room temperature (Fig. 2(a)). The dry and hardened CaP blocks were carefully ground and sieved into micron-sized particles at 0.25 and 1 mm. Scanning electron microscopy (SEM) images revealed that the granules with unique nano-topography exhibited a rough and slightly uneven appearance with nano-HAP depositions (Figs. 3(a)–3(c)). High-resolution transmission electron microscope (TEM) showed that micron-sized granule was comprised of needle-like HAP crystals with the grain length at about 100 nm (Fig. 3(d)).

This novel bone repairing granule was based on the biomimetic calcium phosphorus coating [17]. Wu et al. deposited organic polymers with a calcium phosphate layer using this biomimetic method [46]. Zheng et al. used a layer-by-layer approach combined with BMP-2 to produce highly efficient osteogenic calcium phosphorus particles instead of preparing calcium-phosphorus coatings on existing scaffold [47]. This is the key shift of biomimetic calcium phosphorus coating technology from a two-dimensional coating to a three-dimensional particle. However, the mechanical properties of these layer-assembled calcium phosphorus particles are not high and cannot be directly used as material bone filling material [48]. By increasing the concentration of key ions in the reaction solution, the bio-bone granules with osteoconduction properties and cancellous-like stiffness could be developed [47, 49, 50].

This advanced bio-bone granule turned nano-sized particles to micron sized particles (1.25–1 mm) which is more suitable for bone-filling materials. It is a carbonated and calcium deficient nano-HAP with crystal defects and an abundance of substitutions (traces of Na^+ and Cl^-) which exhibit a significant variation from the stoichiometric nano-HAP [45]. Therefore, the overall advantages can be summarized as follows: Firstly, the nanoparticles are attracted to each other and condensed at room temperature to form biobone granules without sintering. Moreover, the hardness of this granule is moderate and similar to the mineral phase of human cancellous bone; Secondly, the preparation process can internally incorporate bioactive ingredients within granules which avoided the burst release caused by surface adsorption. Thirdly, the non-stoichiometric nano-HAP aggregated granule is more smartly responsive and conditionally degraded with the bone healing process. In the early stages of osteogenesis, it provides stable space for

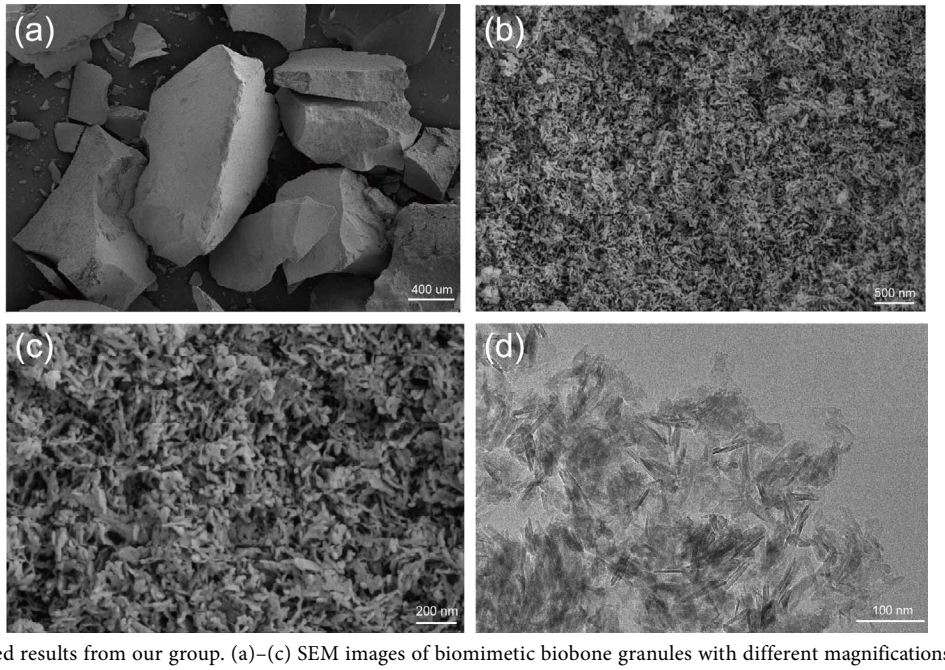

Figure 3 Unpublished results from our group. (a)–(c) SEM images of biomimetic biobone granules with different magnifications; (d) TEM images of nanocrystalline hexagonal hydroxyapatite.

new bone formation, while the material slowly degrades and gradually transforms into new bone tissue in the later stages of bone resorption and remodeling. Finally, it is an eco-friendly, high biosafety, and low energy consumption synthesis method suitable for large-scale industrial production.

5 Nano-HAP prepared at macro sized scaffold

Macroscopically designed nano-HAP scaffold was usually fabricated for osteogenesis and osteochondral tissue-engineering coupled with bioactive factors and stem cells. Zhu et al. prepared a nano-HAP via collosol-flocculation method and made into polyphonous macro scaffold with the wood-model method [51]. They further functionalized nano-HAP scaffold by dropping RhBMP-2 solution on the surface of scaffold [52]. Both preclinical and clinical research had suggested that nano-HAP scaffold could be an ideal material for the reconstruction of bone defects [52, 53]. Smith et al fabricated scaffolds using both nano HAP powder (20 nm average particle size) by sintering and found more osteoblasts proliferation than prepared with micro HA powder (10 μm average particle size) [54]. Low-temperature sintering was also used in nano-HAP scaffolds with the average size of the inner nanopore being about 16.5–30.2 nm [55]. Nowadays, three-dimensional (3D) printing of implants is relatively mature and the incorporation of nano-HAP into biomedical polymers is the most commonly used for printing ink. Lian et al. reported a nano-HAP incorporated 3D biomimetic porous sponge-like scaffold by a low-temperature-printed method and results confirmed MSC-laden this scaffold significantly promoted vascularized in rat distal femoral defect model [56]. Surface coating technology is another strategy to modify 3D-printed scaffolds with biomimetic nano-HAP [57]. Nano-HAP composites scaffold outperform nano-HAP alone in repairing traumatized or disease-affected large bone defects due to its mechanical stability, customizable pore size, and multifunctional modification [58].

6 Application of nano-HAP

Nano-HAP is the most widely studied nanomaterials for various biomedical applications. It is an inorganic composition identical to the bone structure with good biocompatibility and bioactivity. As a grafted material, nano-HAP can be fabricated into macroscopic particles or blocks for the reconstruction of large-scale defects that result from trauma or cancer [53]. And the coating of nano-HAP mineralized layers on bio-inert implants would achieve better osteointegration and functional longevity and prevent recurrence of tumor *in situ* [8]. In tissue engineering, researchers embedded nano-HAP particles into a 3D scaffold combined with stem cells creating a favorable osteogenic microenvironment [56]. An *in vitro* experiment demonstrated the early biological response of stem cells on bone cement conducted by Che and co-workers [59]. It has shown the addition of the nano-HAP phase into PMMA-based bone cement improved cell adhesion, spreading, proliferation, and early osteogenesis. Nano-HAP is a substance biomaterial that improves the osseointegration between implants and host bone, and has multiple excellent bioactive capacities (osteoinductivity, antitumor, antibacterial) [8, 38, 60].

In addition to the above-reported orthopedic application, it is worth mentioning the application of nano-HAP in dentistry. Acidic microenvironments from bacterial metabolism contribute to tooth decay and erosion by dissolving the inorganic mineral phase of teeth, HAP. Unfortunately, mature enamel cannot self-construct with the help of osteoblasts or osteoclasts as the bone healing process do because it is acellular. Therefore, the optimal way for enamel regeneration is to inhibit tooth demineralization or promote tooth remineralization. Nowadays, nano-HA has been incorporated in toothpaste to provide ions helping re-mineralize the remaining collagen network. De Carvalho et al. tested the protective effect of nano-HAP paste on artificially produced on enamel caries lesions and found nano-HAP is more effective than other count parts in Knoop microhardness testing (KHN) [61]. In addition, Nano-HAP can serve as the desensitizing agent. Vano et al. conducted a long-term randomized clinical trial to verify whether 2% nano-HAP enhanced the bleaching effect of 6% hydrogen peroxide [62]. The results revealed that the addition of 2% nano-HAP can prevent hypersensitivity significantly. A recent systematic review and meta-analysis have compared nano-HAP graft and other bone grafts in the repair of periodontal infrabony lesions, which showed that nano-HAP has equivalent effects to other bone grafts in terms of clinical attachment level gain, probing pocket depth decrease, and gingival recession changes [63]. Therefore, nano-HAP may have a great clinical application prospect in periodontal regeneration. In addition, due to its excellent biocompatibility, antibacterial activity, and bone-like composition, nano-HAP has been widely used in implantology, as a coating material coated with implant surface by itself or in combination with other nano-materials to promote osseointegration and improve bone nanomechanical properties [64–66].

7 Conclusions

Here, we discussed the biomimetic preparation method of nano-HAP in biomedical applications and figured out its difference from conventional routes. The initial reactants, physiological reaction conditions, and no extra heat treatment are significant features of biomimetic preparation. This process with the merits of lower energy consumption and higher biosafety is promising for clinical translation (for example mineralized collagen composites) [22]. In addition, the currently available biomimetic synthesis methods allow us to obtain multiple forms of nanocrystalline HAP such as nanoparticles, small granules, and blocks, but few reported the biomimetic precipitation mechanism of crystal growth and specific morphology, controllable size, ions substitution of nano-HAP. The key challenge remains to improve the physiochemical properties and mechanical and biological properties of the nano-HAP composites. This review will provide a useful reference for the preparation, optimization, and practical application of biomimetic nano-HAP.

Author contributions

B. W. drafted the manuscript. Y. R. and C. S. summarized related publications. G. W. and H. L. designed the work. G. X. prepared the SEM image of biobone granules. H. L., T. F., and G. W. reviewed the manuscript. All authors read and approved the submitted version.

Declaration of conflicting interests

The authors declare no conflict of interests.

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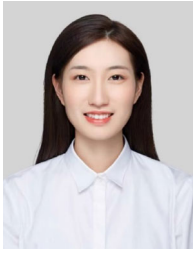

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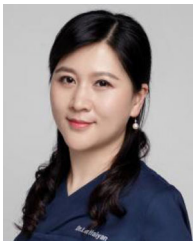

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